

CO2xNxPhrag: Porewater Methane

May 2026 Samples

2026-07-21

Contents

1	Read in data files	2
2	Assess standard curves	2
3	Anlayze the Check Standards	5
4	Analyze the Blanks	6
5	Dilution correct samples and Check Range	9
6	Calculate gas in water	10
7	Merge samples with Metadata & check all samples are present	10

##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

Run Information: Brenden D. Blakley

```
#set the run date & user name
```

```
run_date <- "06/12/2026"
```

```
sample_year <- "2026"
```

```
sample_month <- "May"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
GHG_files <- c("Raw Data/20260612_GHGdata_LTREB_Phrag.csv")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- "Processed Data/LTREB_Phrag-ShimadzuGC-QAQC_2026.csv"
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/GCReW_CO2xNxPhrag_Porewater_CH4_202505.csv"
```

```
#Gas in Water Calculations
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- ""
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

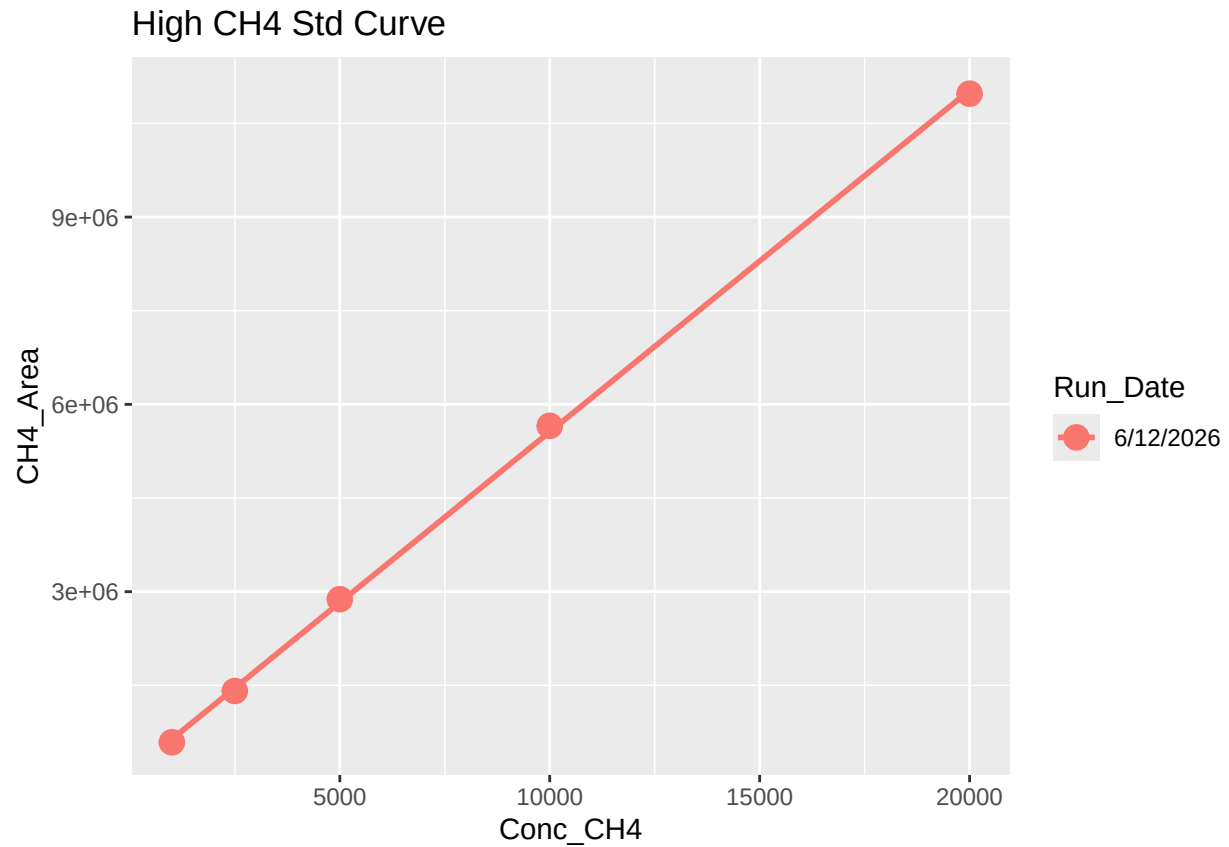
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

1 Read in data files

2 Assess standard curves

'geom_smooth()' using formula = 'y ~ x'



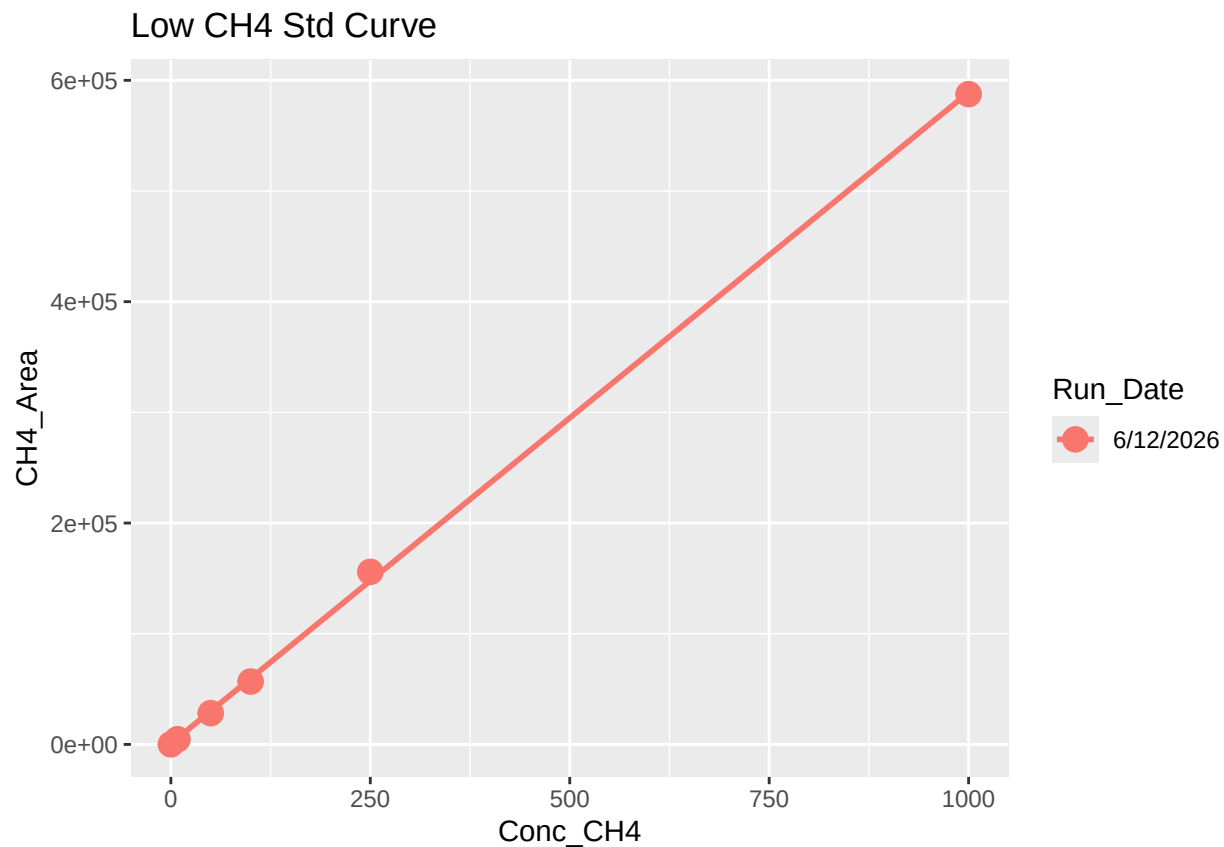
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## ('geom_point()').
```

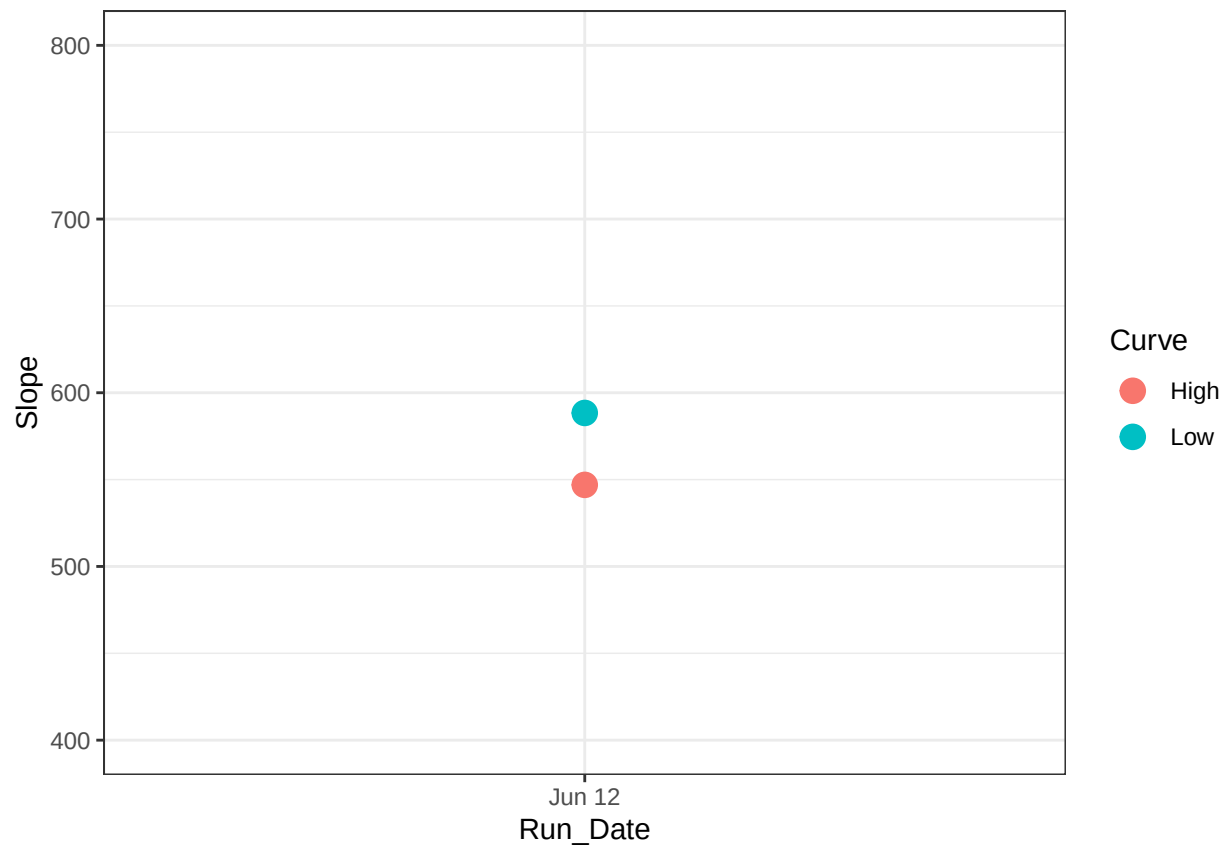


```
## Joining with 'by = join_by(Run_Date)'
```

```
##   X Curve      R2    Slope  Intercept  Run_Date
## 1 1  High 0.9997391 577.5898 104642.5944 2024-09-02
## 2 2  High 0.9998755 586.6350  48939.9343 2025-09-05
## 3 3   Low 0.9997620 599.1783    606.0063 2024-09-02
## 4 4   Low 0.9999368 601.4464    195.9643 2025-09-05
## 5 5  High 0.9999761 599.9888   21668.5842 2025-06-23
## 6 6  High 0.9991361 578.3234 147368.3056 2025-06-24
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Run_Date = mdy(Run_Date)'.
## Caused by warning:
## ! 17 failed to parse.
```

```
## Warning: Removed 17 rows containing missing values or values outside the scale range
## ('geom_point()').
```



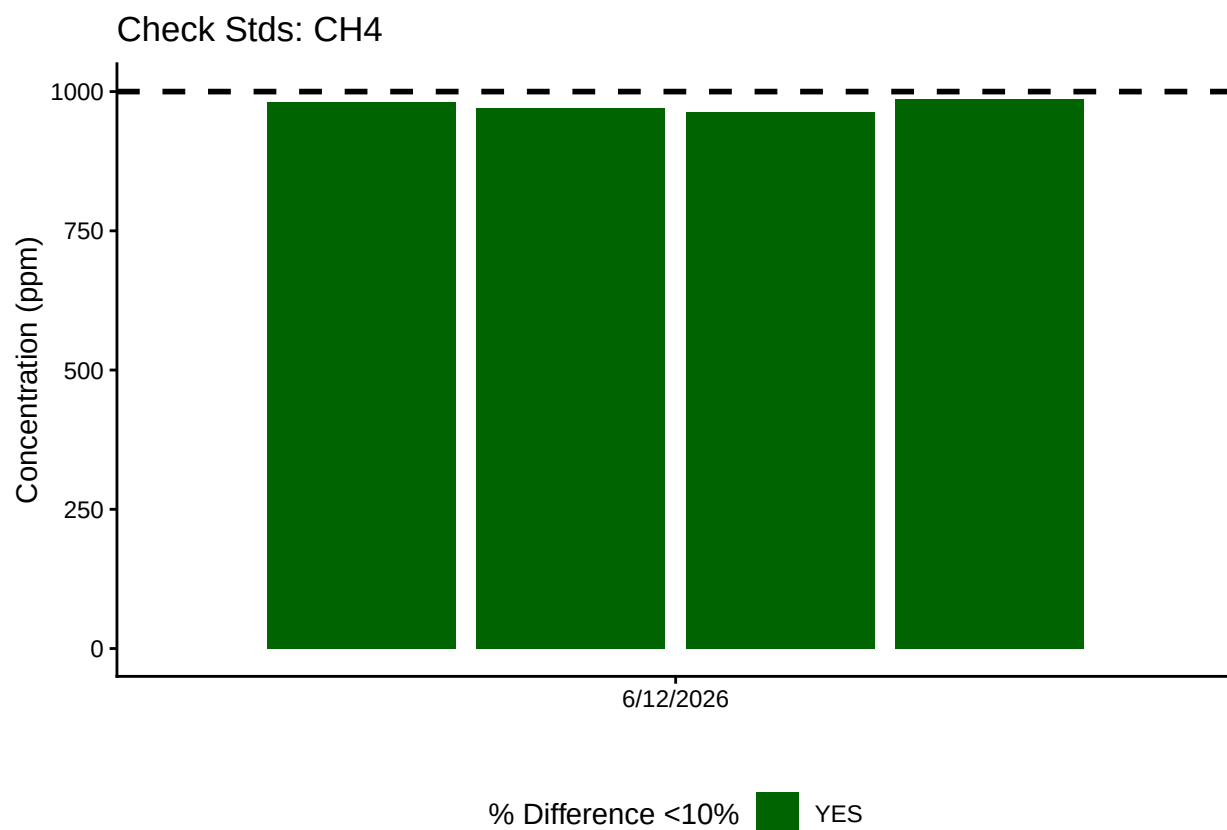
```
## [1] "High_R2 is Good-6/12/2026"
```

```
## [1] "Low_R2 is Good-6/12/2026"
```

2.1 Now calculate the CH₄ & CO₂ concentrations in ppm

3 Anlayze the Check Standards

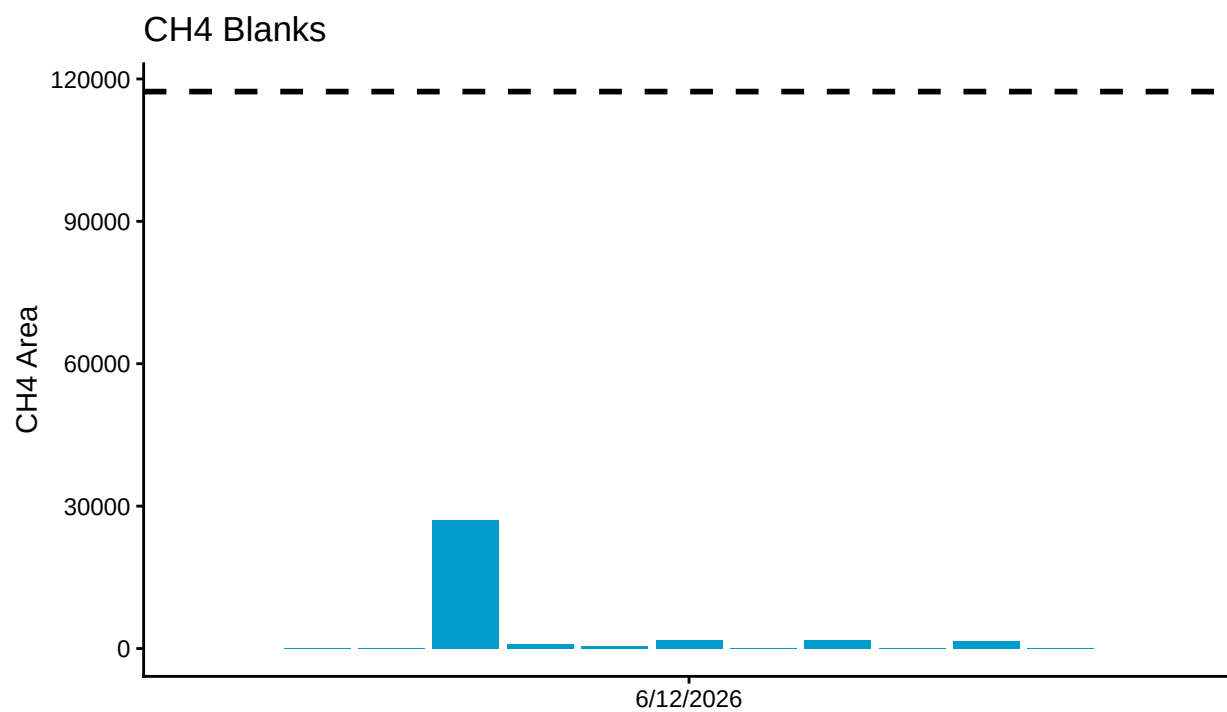
```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

Assess Blanks

>60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

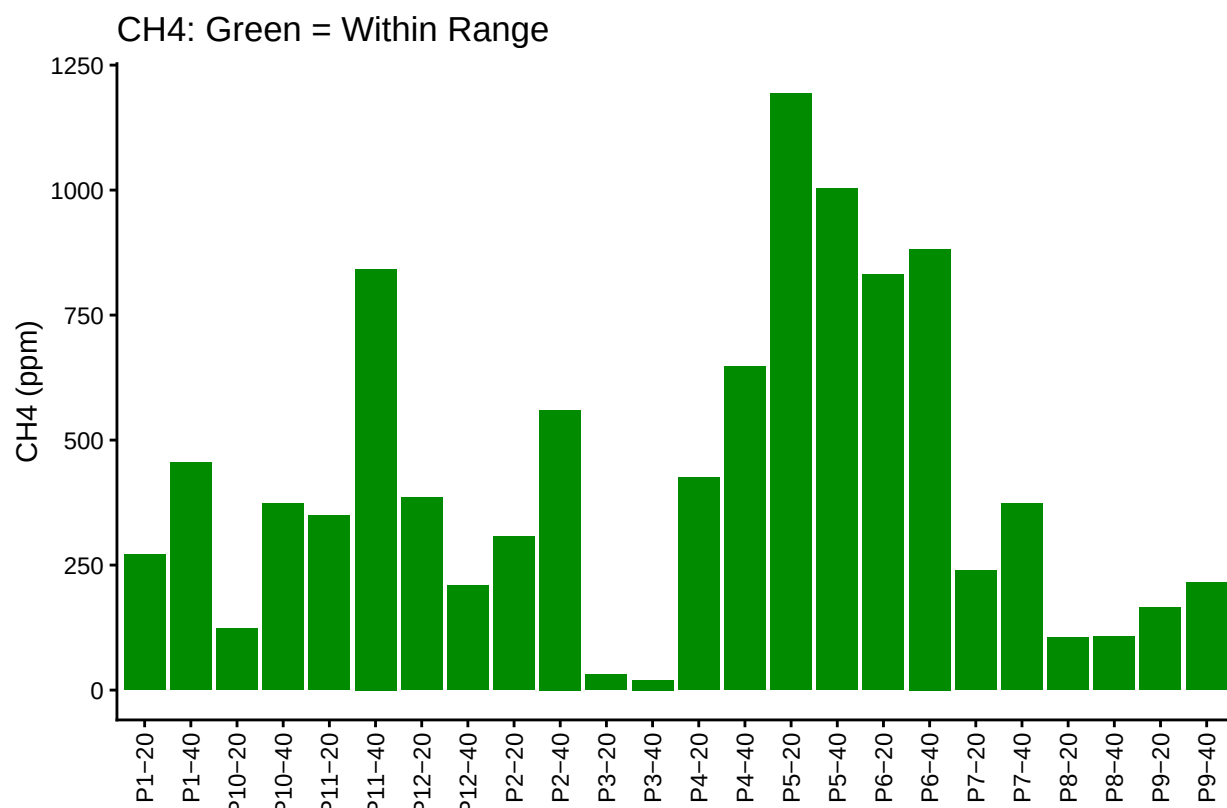
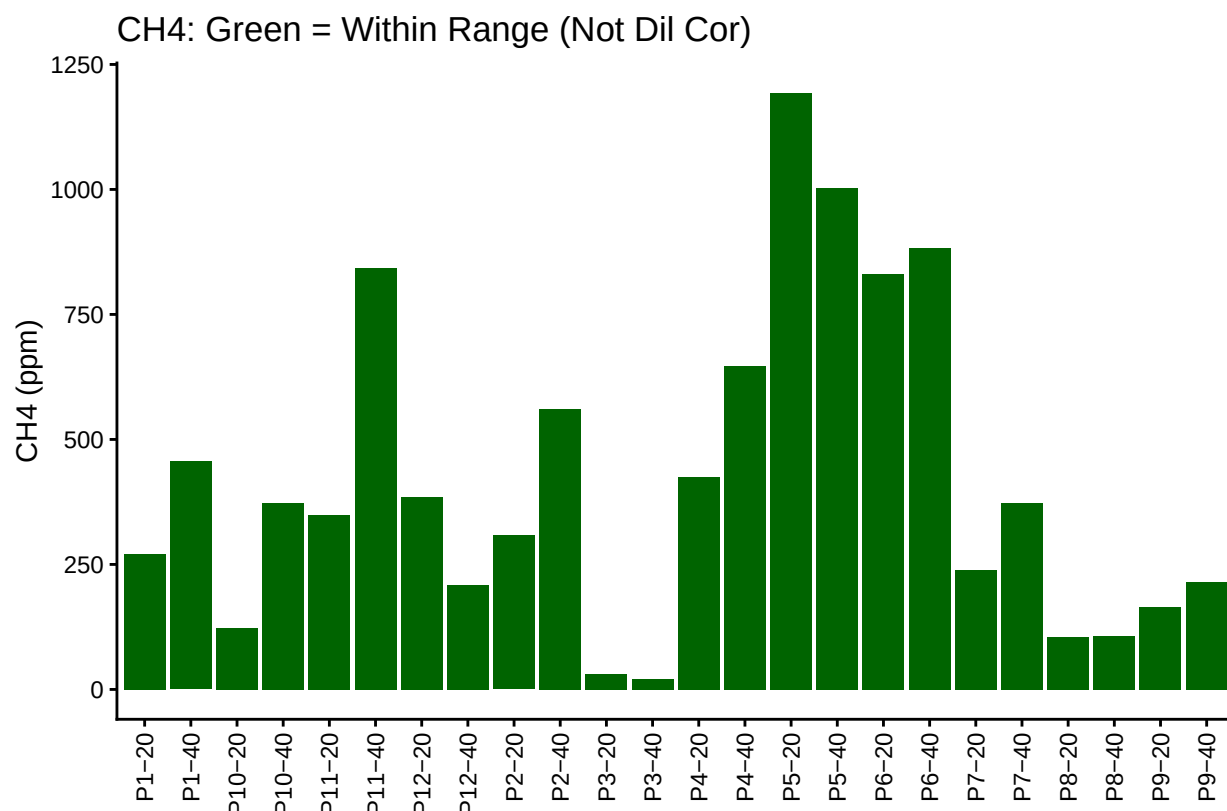


Blank Conc <25% Quartile Samples ■ YES

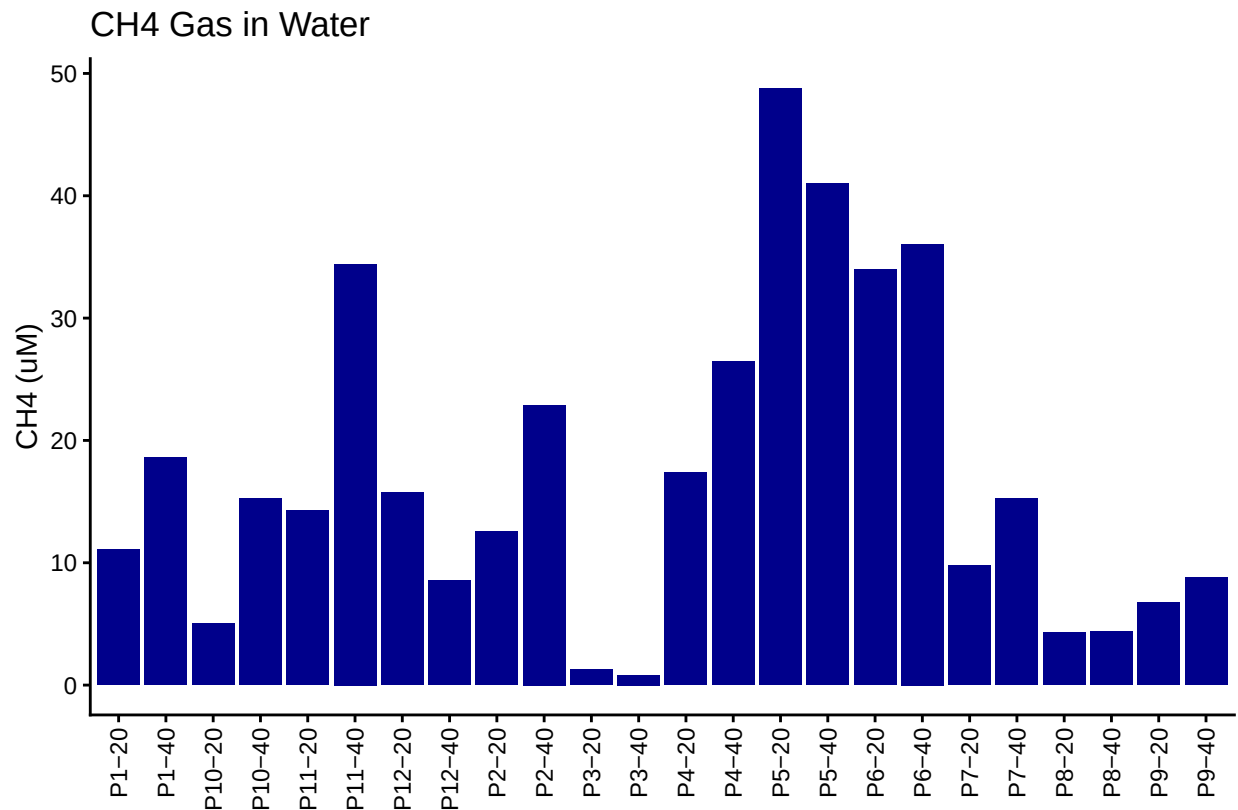
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	3125.091

5 Dilution correct samples and Check Range



6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

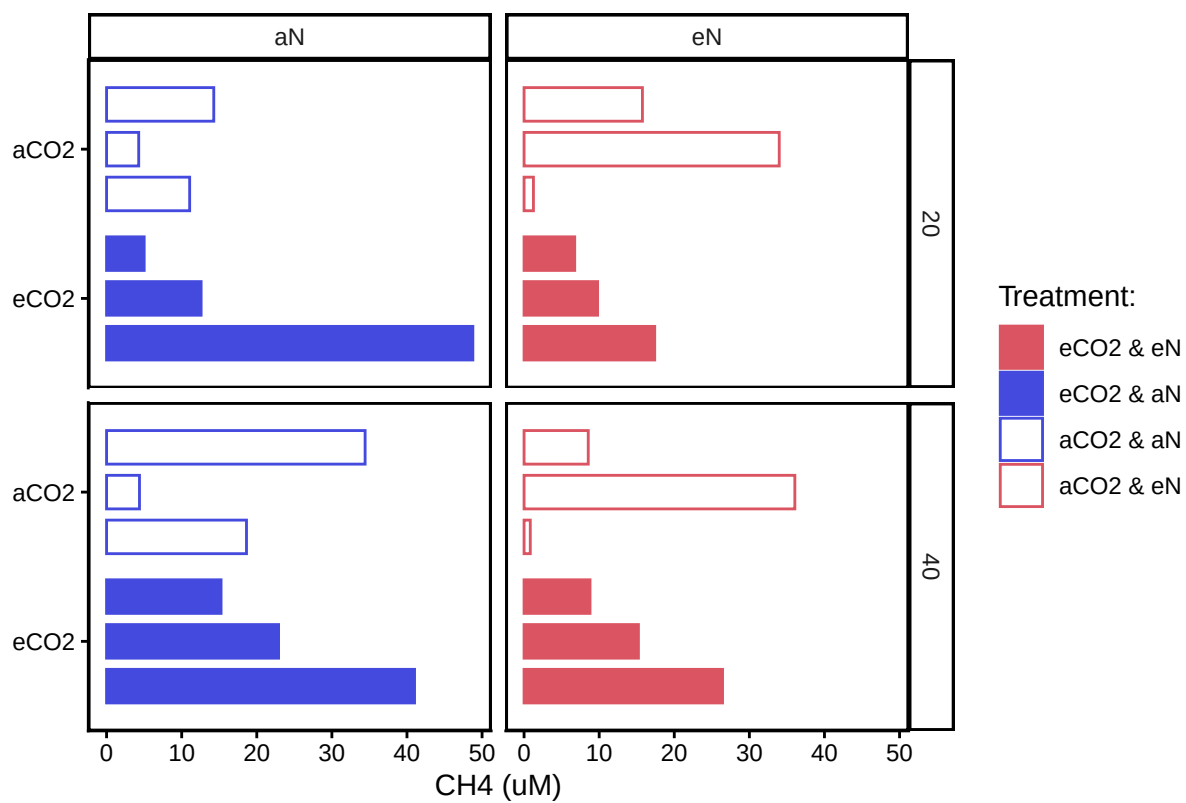
```
## All sample IDs are present in data
```

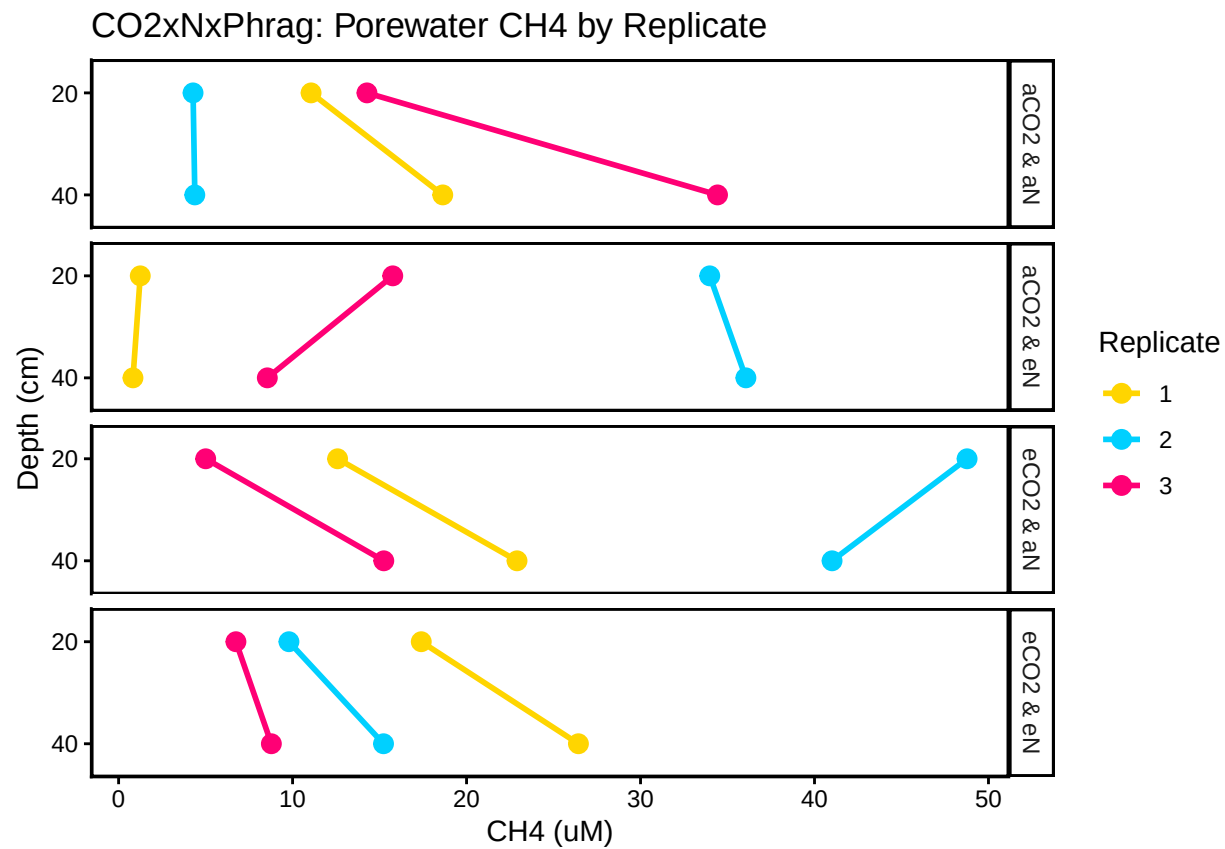
7.1 Organize Data

7.2 Visualize Data

```
## Visualize Data
```

CO2xNxPhrag: Porewater Methane





#Export Processed Data

```
write.csv(All_Clean_Data, final_path)
```

#end